

Homework 10

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Problem 10.1.

Solution 10.1.

1. Since all columns of Q are perpendicular, to make each column of Q is ONB, their length should be 1, so $k = \pm \frac{1}{2}$.
2. Q is a symmetric matrix and all the columns are mutually perpendicular, so

$$Q^T Q = k^2 \cdot 4I_4 \implies \frac{1}{4k^2} Q^T Q = I_4 \implies Q^{-1} = \frac{1}{4k^2} Q^T = \frac{1}{4k^2} Q$$

3. The orthogonal projection matrix P to the first column is $P = \frac{1}{4} \begin{bmatrix} 1 & -1 & -1 & -1 \\ -1 & 1 & 1 & 1 \\ -1 & 1 & 1 & 1 \\ -1 & 1 & 1 & 1 \end{bmatrix}$, so the projection is

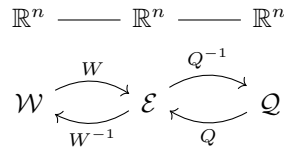
$$\frac{1}{4} \begin{bmatrix} 1 & -1 & -1 & -1 \\ -1 & 1 & 1 & 1 \\ -1 & 1 & 1 & 1 \\ -1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -\frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \end{bmatrix}.$$

4. The orthogonal projection matrix to the first two columns is $\begin{bmatrix} \frac{1}{2} & \frac{-1}{2} & 0 & 0 \\ \frac{-1}{2} & \frac{1}{2} & 0 & 0 \\ 0 & 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & \frac{1}{2} & \frac{1}{2} \end{bmatrix}$, so the projection is

$$\begin{bmatrix} \frac{1}{2} & \frac{-1}{2} & 0 & 0 \\ \frac{-1}{2} & \frac{1}{2} & 0 & 0 \\ 0 & 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & \frac{1}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}.$$

5. They are orthonormal.

Let basis $\mathcal{W}$ is the columns of matrix W , basis $\mathcal{Q}$ is the columns of matrix Q , and basis $\mathcal{E}$ is the standard basis $\mathcal{E} = (e_1, e_2, \dots, e_n)$. Then the change of coordinates maps between these three basis can be illustrate as the following graph.



That is to say, the change of coordinate matrix from basis $\mathcal{W}$ to $\mathcal{E}$ is W , and the change of coordinate matrix from basis $\mathcal{Q}$ to $\mathcal{E}$ is Q (which is easy to prove.) So,

$$I_{\mathcal{W} \leftarrow \mathcal{Q}} = W^{-1}Q = W^T Q = \begin{bmatrix} \frac{-1}{2} & \frac{-1}{2} & \frac{-1}{2} & \frac{-1}{2} \\ \frac{1}{2} & \frac{1}{2} & \frac{-1}{2} & \frac{-1}{2} \\ \frac{\sqrt{2}}{2} & \frac{-\sqrt{2}}{2} & 0 & 0 \\ 0 & 0 & \frac{\sqrt{2}}{2} & \frac{-\sqrt{2}}{2} \end{bmatrix}.$$

$$I_{\mathcal{Q} \leftarrow \mathcal{W}} = Q^{-1}W = Q^T W = \begin{bmatrix} \frac{-1}{2} & \frac{1}{2} & \frac{\sqrt{2}}{2} & 0 \\ \frac{-1}{2} & \frac{1}{2} & \frac{-\sqrt{2}}{2} & 0 \\ \frac{-1}{2} & \frac{-1}{2} & 0 & \frac{\sqrt{2}}{2} \\ \frac{-1}{2} & \frac{-1}{2} & 0 & \frac{-\sqrt{2}}{2} \end{bmatrix}.$$

Problem 10.2.

Solution 10.2.

1. Set $\mathbf{a}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $\mathbf{a}_2 = \begin{bmatrix} 4 \\ 0 \end{bmatrix}$. It's obvious that they are not ONB, let's do GS Orthogonalization. First, we do dot product with all the vectors with $\mathbf{a}_1$.

$$\text{dot product with } \mathbf{a}_1 \quad \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \begin{bmatrix} 4 \\ 0 \end{bmatrix}$$

Then we use the first column to kill the dot product by column operations.

$$\hat{\mathbf{q}}_1 = \mathbf{a}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\hat{\mathbf{q}}_2 = \mathbf{a}_2 - 2\mathbf{a}_1 = \begin{bmatrix} 2 \\ -2 \end{bmatrix}$$

Now $\hat{\mathbf{q}}_1, \hat{\mathbf{q}}_2$ are perpendicular.

$$A = \hat{Q}\hat{R} = \begin{bmatrix} 1 & 2 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix}$$

Finally, we divided each columns by its length such that they are length-one. (Normalization)

$$\begin{aligned} A &= \begin{bmatrix} 1 & 2 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix} \\ &= \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{2}{2\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{2}{2\sqrt{2}} \end{bmatrix} \begin{bmatrix} \sqrt{2} & \\ & 2\sqrt{2} \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix} \\ &= \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \sqrt{2} & 2\sqrt{2} \\ & 2\sqrt{2} \end{bmatrix} = QR \end{aligned}$$

The orthogonal projection matrix to its range is

$$P = QQ^T = I_2$$

2. Similar to 1,

$$\begin{aligned} A &= \hat{Q}\hat{R} = \begin{bmatrix} 1 & 2 \\ 2 & 1 \\ -2 & 2 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \\ &= \begin{bmatrix} \frac{1}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{1}{3} \\ -\frac{2}{3} & \frac{2}{3} \end{bmatrix} \begin{bmatrix} 3 & \\ & 3 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \\ &= \begin{bmatrix} \frac{1}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{1}{3} \\ -\frac{2}{3} & \frac{2}{3} \end{bmatrix} \begin{bmatrix} 3 & -3 \\ & 3 \end{bmatrix} = QR \end{aligned}$$

The orthogonal projection matrix to its range is

$$P = QQ^T = \begin{bmatrix} \frac{5}{9} & \frac{4}{9} & \frac{2}{9} \\ \frac{4}{9} & \frac{5}{9} & -\frac{2}{9} \\ \frac{2}{9} & -\frac{2}{9} & \frac{8}{9} \end{bmatrix}$$

3. Set $\mathbf{a}_1 = \begin{bmatrix} 2 \\ 1 \\ 0 \\ 1 \end{bmatrix}$, $\mathbf{a}_2 = \begin{bmatrix} 0 \\ 1 \\ 2 \\ 2 \end{bmatrix}$, $\mathbf{a}_3 = \begin{bmatrix} -1 \\ 1 \\ 1 \\ 2 \end{bmatrix}$. It's obvious that they are not ONB, let's do GS Orthogonalization.

First, we do dot product with all the vectors with $\mathbf{a}_1$.

$$\text{dot product with } \mathbf{a}_1 \quad \begin{bmatrix} 2 \\ 1 \\ 0 \\ 1 \end{bmatrix} \quad \begin{bmatrix} 0 \\ 1 \\ 2 \\ 2 \end{bmatrix} \quad \begin{bmatrix} -1 \\ 1 \\ 1 \\ 2 \end{bmatrix}$$

Then we use the first column to kill the dot product by column operations.

$$\hat{\mathbf{q}}_1 = \mathbf{a}_1 = \begin{bmatrix} 2 \\ 1 \\ 0 \\ 1 \end{bmatrix}$$

$$\hat{\mathbf{q}}_2 = \mathbf{a}_2 - \frac{1}{2}\mathbf{a}_1 = \begin{bmatrix} -1 \\ \frac{1}{2} \\ 2 \\ \frac{3}{2} \end{bmatrix}$$

$$\hat{\mathbf{q}}_3 = \mathbf{a}_3 - \frac{1}{6}\mathbf{a}_1 = \begin{bmatrix} -\frac{8}{6} \\ \frac{5}{6} \\ 1 \\ \frac{11}{6} \end{bmatrix}$$

Now, $\hat{\mathbf{q}}_1$ is perpendicular to $\hat{\mathbf{q}}_2, \hat{\mathbf{q}}_3$, but $\hat{\mathbf{q}}_2$ and $\hat{\mathbf{q}}_3$ are still not perpendicular. we do dot product with all the vectors except $\hat{\mathbf{q}}_1$ with $\hat{\mathbf{q}}_2$.

$$\begin{array}{ccc} \begin{bmatrix} 2 \\ 1 \\ 0 \\ 1 \end{bmatrix} & \begin{bmatrix} -1 \\ \frac{1}{2} \\ 2 \\ \frac{3}{2} \end{bmatrix} & \begin{bmatrix} -\frac{8}{6} \\ \frac{5}{6} \\ 1 \\ \frac{11}{6} \end{bmatrix} \\ \text{dot product with } \hat{\mathbf{q}}_2 & - & \frac{15}{2} \end{array}$$

Then we use the second column to kill the dot product by column operations.

$$\hat{\mathbf{q}}_3 = \hat{\mathbf{q}}_3 - \frac{13}{15}\hat{\mathbf{q}}_2 = \begin{bmatrix} -\frac{8}{6} \\ \frac{5}{6} \\ 1 \\ \frac{11}{6} \end{bmatrix} - \frac{13}{15} \begin{bmatrix} -1 \\ \frac{1}{2} \\ 2 \\ \frac{3}{2} \end{bmatrix} = \begin{bmatrix} -\frac{7}{15} \\ \frac{2}{15} \\ \frac{5}{15} \\ \frac{8}{15} \end{bmatrix}$$

They are mutually perpendicular now.

$$A = \hat{Q}\hat{R} = \begin{bmatrix} 2 & -1 & -\frac{7}{15} \\ 1 & \frac{1}{2} & \frac{2}{15} \\ 0 & 2 & -\frac{11}{15} \\ 1 & \frac{3}{2} & \frac{8}{15} \end{bmatrix} \begin{bmatrix} 1 & \frac{1}{2} & \frac{1}{15} \\ & 1 & \frac{13}{15} \\ & & 1 \end{bmatrix}$$

Finally, we divided each columns by its length such that they are length-one. (Normalization)

$$\begin{aligned} A &= \begin{bmatrix} \frac{2}{\sqrt{6}} & -\frac{2}{\sqrt{30}} & -\frac{7}{3\sqrt{30}} \\ \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{30}} & \frac{2}{3\sqrt{30}} \\ 0 & \frac{4}{\sqrt{30}} & -\frac{11}{3\sqrt{30}} \\ \frac{1}{\sqrt{6}} & \frac{3}{\sqrt{30}} & \frac{8}{3\sqrt{30}} \end{bmatrix} \begin{bmatrix} \sqrt{6} & & \\ & \frac{\sqrt{30}}{2} & \\ & & \frac{\sqrt{30}}{5} \end{bmatrix} \begin{bmatrix} 1 & \frac{1}{2} & \frac{1}{15} \\ & 1 & \frac{13}{15} \\ & & 1 \end{bmatrix} \\ &= \begin{bmatrix} \frac{2}{\sqrt{6}} & -\frac{2}{\sqrt{30}} & -\frac{7}{3\sqrt{30}} \\ \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{30}} & \frac{2}{3\sqrt{30}} \\ 0 & \frac{4}{\sqrt{30}} & -\frac{11}{3\sqrt{30}} \\ \frac{1}{\sqrt{6}} & \frac{3}{\sqrt{30}} & \frac{8}{3\sqrt{30}} \end{bmatrix} \begin{bmatrix} \sqrt{6} & \frac{\sqrt{6}}{2} & \frac{\sqrt{6}}{6} \\ & \frac{\sqrt{30}}{2} & \frac{13\sqrt{30}}{30} \\ & & \frac{\sqrt{30}}{5} \end{bmatrix} = QR \end{aligned}$$

The orthogonal projection matrix to its range is

$$P = QQ^T = \begin{bmatrix} \frac{53}{54} & \frac{1}{9} & \frac{1}{54} & -\frac{2}{27} \\ \frac{1}{9} & \frac{1}{3} & -\frac{1}{9} & \frac{2}{9} \\ \frac{1}{54} & -\frac{1}{9} & \frac{53}{54} & \frac{2}{27} \\ -\frac{2}{27} & \frac{2}{9} & \frac{2}{27} & \frac{19}{27} \end{bmatrix}.$$

4. Similar to 3, the QR decomposition of A is

$$A = \begin{bmatrix} \frac{1}{\sqrt{6}} & \frac{4}{\sqrt{30}} & -\frac{4}{\sqrt{70}} \\ \frac{\sqrt{6}}{2} & -\frac{1}{\sqrt{30}} & \frac{3\sqrt{70}}{14} \\ -\frac{\sqrt{6}}{3} & \frac{\sqrt{30}}{3} & \frac{3\sqrt{70}}{11} \\ 0 & \frac{3}{\sqrt{30}} & \frac{3}{\sqrt{70}} \end{bmatrix} \begin{bmatrix} \sqrt{6} & -\frac{\sqrt{6}}{3} & \frac{\sqrt{6}}{3} \\ \frac{\sqrt{30}}{3} & -\frac{\sqrt{30}}{3} & \frac{3\sqrt{30}}{10} \end{bmatrix} = QR$$

The orthogonal projection matrix to its range is

$$P = QQ^T = \begin{bmatrix} \frac{13}{14} & -\frac{1}{21} & -\frac{1}{6} & \frac{4}{21} \\ \frac{1}{21} & \frac{61}{63} & -\frac{1}{9} & \frac{63}{4} \\ -\frac{1}{6} & -\frac{1}{9} & \frac{11}{18} & \frac{9}{31} \\ \frac{4}{21} & \frac{8}{63} & \frac{4}{9} & \frac{63}{63} \end{bmatrix}$$

Problem 10.3.

Solution 10.3.

1. The ONB of the kernel is $\frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$, the projection matrix is $\begin{bmatrix} \frac{5}{6} & \frac{1}{6} & \frac{1}{3} \\ \frac{1}{6} & \frac{2}{3} & -\frac{1}{3} \\ \frac{1}{3} & -\frac{1}{3} & \frac{1}{3} \end{bmatrix}$.

2. Projection matrix $\begin{bmatrix} \frac{1}{6} & -\frac{1}{6} & -\frac{1}{3} \\ -\frac{1}{6} & \frac{1}{6} & \frac{2}{3} \\ -\frac{1}{3} & \frac{2}{3} & \frac{2}{3} \end{bmatrix}$.

3. Projection matrix $\frac{1}{25} \begin{bmatrix} 9 & 12 \\ 12 & 16 \end{bmatrix}$.

4. The ONB of $\text{Ran}(A^T)$ is $\frac{1}{9} \begin{bmatrix} 3 \\ 6 \\ 6 \end{bmatrix}$, so the orthogonal projection matrix to the $\text{Ran}(A^T)$ is $\frac{1}{9} \begin{bmatrix} 1 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 4 \end{bmatrix}$. Since

$\ker(A) = \text{Ran}(A^T)^\perp$, the orthogonal projection to $\ker(A)$ is $\begin{bmatrix} 1 & & \\ & 1 & \\ & & 1 \end{bmatrix} - \frac{1}{9} \begin{bmatrix} 1 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 4 \end{bmatrix} = \frac{1}{9} \begin{bmatrix} 8 & -2 & -2 \\ -2 & 5 & -4 \\ -2 & -4 & 5 \end{bmatrix}$.

5. Projection matrix $\frac{1}{6} \begin{bmatrix} 5 & 2 & -1 \\ 2 & 2 & 2 \\ -1 & 2 & 5 \end{bmatrix}$.

Problem 10.4.

Solution 10.4.

1. Proof:

Set the i -th column of P is $\mathbf{p}_i$, its length squared is $\mathbf{p}_i^T \mathbf{p}_i$, then

$$\mathbf{p}_i^T \mathbf{p}_i = (P\mathbf{e}_i)^T (P\mathbf{e}_i) = \mathbf{e}_i^T P^T P \mathbf{e}_i = \mathbf{e}_i^T P \mathbf{e}_i$$

which is the (i, i) entry of P .

2. Proof:

First, we prove that if $AB = BA$, then AB is an orthogonal projection matrix.

Note that A, B are both orthogonal projection matrix, and $AB = BA$, then

$$(AB)^2 = ABAB = AAB B = AB$$

$$(AB)^T = B^T A^T = BA = AB.$$

So AB is an orthogonal projection matrix.

Next, we prove that if AB is an orthogonal projection matrix, then $AB = BA$.

AB is an orthogonal projection matrix $\implies (AB)^T = AB \implies B^T A^T = AB \implies BA = AB$.

Thus, $AB = BA$ if and only if AB is an orthogonal projection matrix.

AB is an orthogonal projection to $\text{Ran}(A) \cap \text{Ran}(B)$.

Proof:

AB is a projection to $\text{Ran}(AB)$. Now we prove that $\text{Ran}(AB) = \text{Ran}(A) \cap \text{Ran}(B)$.

$\forall \mathbf{x} \in \mathbb{R}^n, AB\mathbf{x} = A(B\mathbf{x}) \in \text{Ran}(A)$, but at the same time, $AB\mathbf{x} = BA\mathbf{x} = B(A\mathbf{x}) \in \text{Ran}(B)$. So

$$\text{Ran}(AB) \subseteq \text{Ran}(A) \cap \text{Ran}(B) \quad (1)$$

$\forall \mathbf{x} \in \text{Ran}(A) \cap \text{Ran}(B), AB\mathbf{v} = A\mathbf{v} = \mathbf{v} \implies \mathbf{v} \in \text{Ran}(AB)$. Hence

$$\text{Ran}(A) \cap \text{Ran}(B) \subseteq \text{Ran}(AB) \quad (2)$$

Combine (1)(2), we have $\text{Ran}(A) \cap \text{Ran}(B) = \text{Ran}(AB)$.

3.

Proof:

If A, B are orthogonal projection matrix, then $\text{Ran}(A) \perp \text{Ran}(B)$ if and only if $AB = BA = O$. So we only need to prove $A + B$ is an orthogonal projection if and only if $AB = BA = O$.

First, we prove $AB = BA = O \implies A + B$ is an orthogonal projection.

Note that A, B are orthogonal projection matrix, then

$$(A + B)^T = A + B$$

$$(A + B)^2 = A^2 + B^2 + AB + BA = A + B + AB + BA = A + B$$

So if $AB = BA = O$, then $A + B$ is an orthogonal projection.

Next, we prove $A + B$ is an orthogonal projection $\implies AB = BA = O$.

If $A + B$ is an orthogonal projection, then

$$(A + B)^2 = A + B \implies A^2 + B^2 + AB + BA = A + B \implies AB + BA = O \implies AB = -BA \quad (3)$$

Right multiply B simultaneously on Equation (3), we have

$$AB^2 + BAB = O \implies AB + BAB = O \quad (4)$$

Combine (3) and (4), we have

$$AB + BAB = O \implies AB + B(-BA) = O \implies AB - BA = O \implies AB = BA.$$

So $AB + BA = O \implies 2AB = O \implies AB = BA = O$.

Then $A + B$ is an orthogonal projection if and only if $AB = BA = O$ if and only if $\text{Ran}(A) \perp \text{Ran}(B)$.

$A + B$ is an orthogonal projection to $\text{Ran}(A) + \text{Ran}(B)$.

Proof:

$A + B$ is a projection to $\text{Ran}(A + B)$. Now we prove that $\text{Ran}(A + B) = \text{Ran}(A) + \text{Ran}(B)$.

$\forall \mathbf{x} \in \mathbb{R}^n, (A + B)\mathbf{x} \in \text{Ran}(A + B)$, at the same time, $(A + B)\mathbf{x} = A\mathbf{x} + B\mathbf{x} \in \text{Ran}(A) + \text{Ran}(B)$. So

$$\text{Ran}(A + B) \subseteq \text{Ran}(A) + \text{Ran}(B) \quad (5)$$

$\forall \mathbf{v} \in \text{Ran}(A) + \text{Ran}(B), \mathbf{v} = \mathbf{a} + \mathbf{b}$, where $\mathbf{a} \in \text{Ran}(A), \mathbf{b} \in \text{Ran}(B)$, then $(A + B)\mathbf{v} = (A + B)(\mathbf{a} + \mathbf{b}) = A\mathbf{a} + A\mathbf{b} + B\mathbf{a} + B\mathbf{b} = \mathbf{a} + \mathbf{b} = \mathbf{v}$, which means $\mathbf{v} \in \text{Ran}(A + B)$. Thus,

$$\text{Ran}(A) + \text{Ran}(B) \subseteq \text{Ran}(A + B) \quad (6)$$

From (5) and (6), we have

$$\text{Ran}(A + B) = \text{Ran}(A) + \text{Ran}(B).$$

So $A + B$ is an orthogonal projection to $\text{Ran}(A) + \text{Ran}(B)$.

Problem 10.4.

Solution 10.4.

1. -6.

2. By row shearing,

$$\begin{vmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 4 & 8 \\ 1 & 3 & 9 & 27 \\ 1 & -1 & 1 & -1 \end{vmatrix} = \begin{vmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 3 & 7 \\ 0 & 2 & 8 & 26 \\ 0 & -2 & 0 & -2 \end{vmatrix} = \begin{vmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 3 & 7 \\ 0 & 0 & 2 & 12 \\ 0 & 0 & 6 & 12 \end{vmatrix} = \begin{vmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 3 & 7 \\ 0 & 0 & 2 & 12 \\ 0 & 0 & 0 & -24 \end{vmatrix} = -48.$$

3. Set $A = \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix}$, $B = \begin{bmatrix} 3 & 4 \\ 4 & 1 \end{bmatrix}$, then

$$\begin{aligned}
 \begin{vmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \end{vmatrix} &= \begin{vmatrix} A & B \\ B & A \end{vmatrix} \\
 &= \begin{vmatrix} A+B & B+A \\ B & A \end{vmatrix} \\
 &= \begin{vmatrix} A+B & O \\ B & A-B \end{vmatrix} \\
 &= \det(A+B) \det(A-B) \\
 &= \begin{vmatrix} 4 & 6 \\ 6 & 4 \end{vmatrix} \cdot \begin{vmatrix} -2 & -2 \\ -2 & 2 \end{vmatrix} \\
 &= (-20) \cdot (-8) \\
 &= 160
 \end{aligned}$$

4. The last three rows must be linearly dependent, so the matrix is not invertible. The determinant is 0.

5. If n is odd, and $\frac{n-1}{2}$ is even, the determinant is 1. If n is odd, and $\frac{n-1}{2}$ is odd, the determinant is -1 . If n is even, and $\frac{n}{2}$ is even, the determinant is 1. If n is even, and $\frac{n}{2}$ is odd, the determinant is -1 .